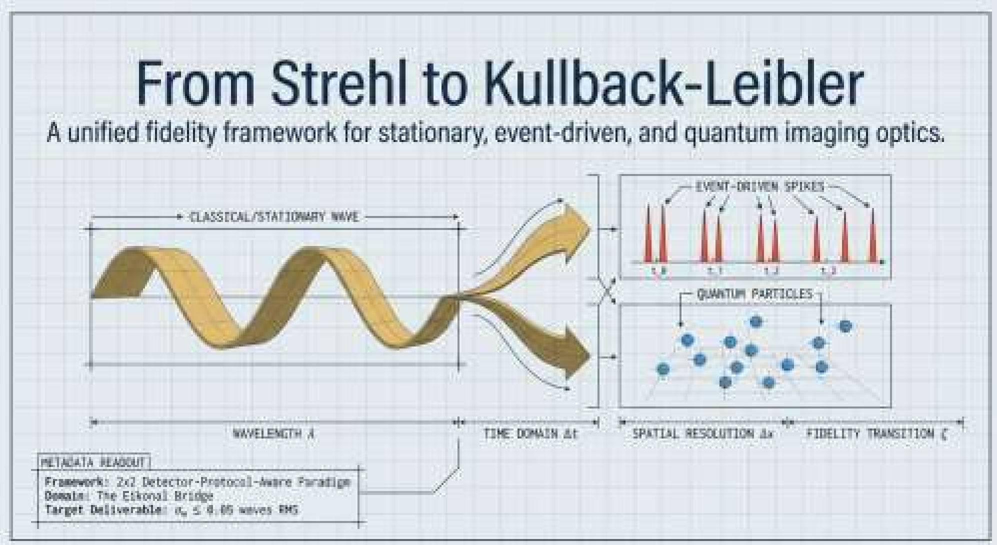
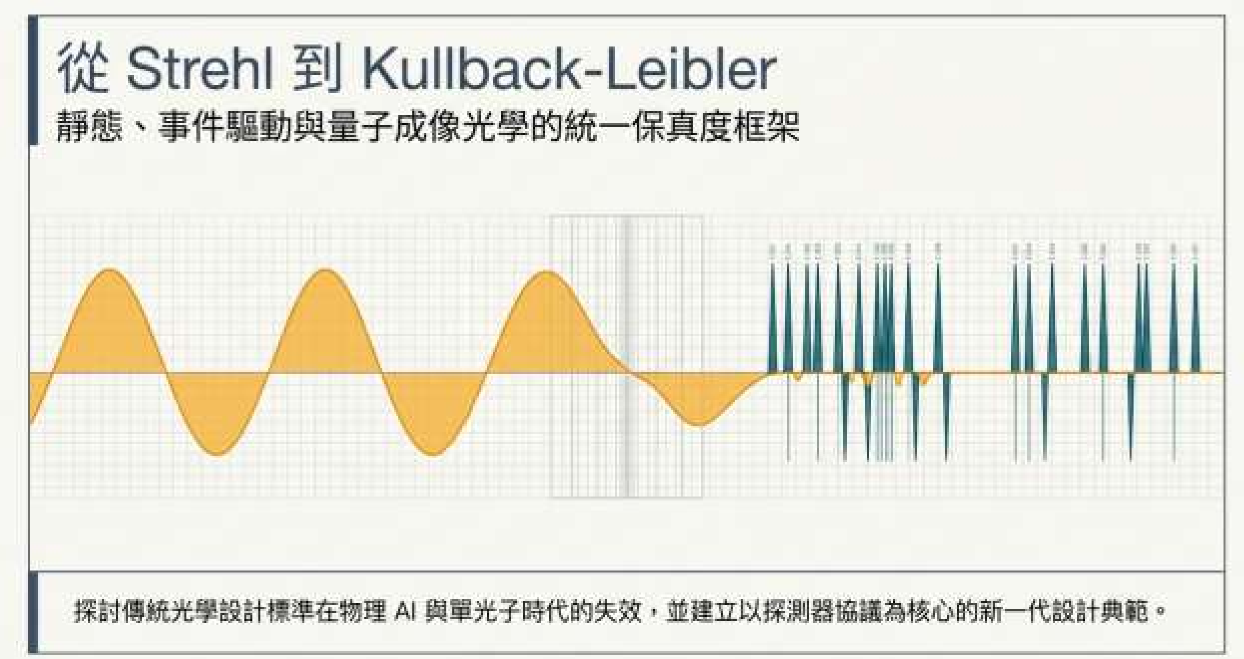
From Strehl to Kullback–Leibler

A detector-protocol-aware lens design playbook for frame, event, and quantum imaging

Engineering seminar deck

For optical designers

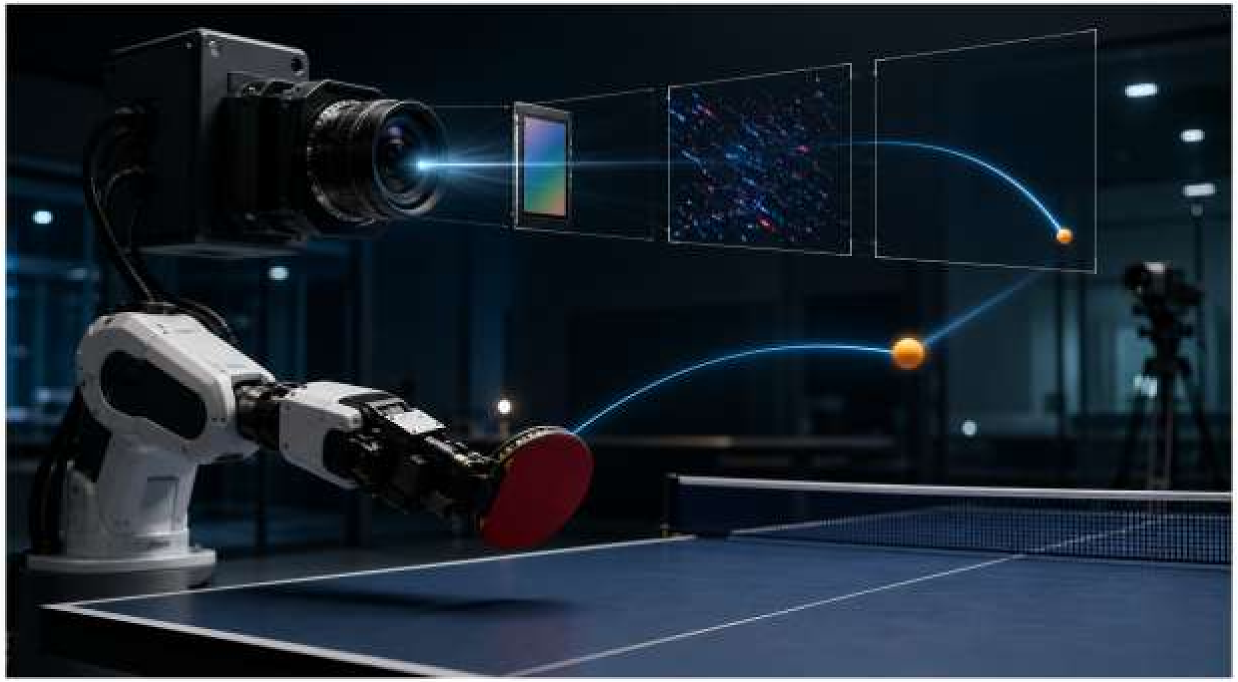


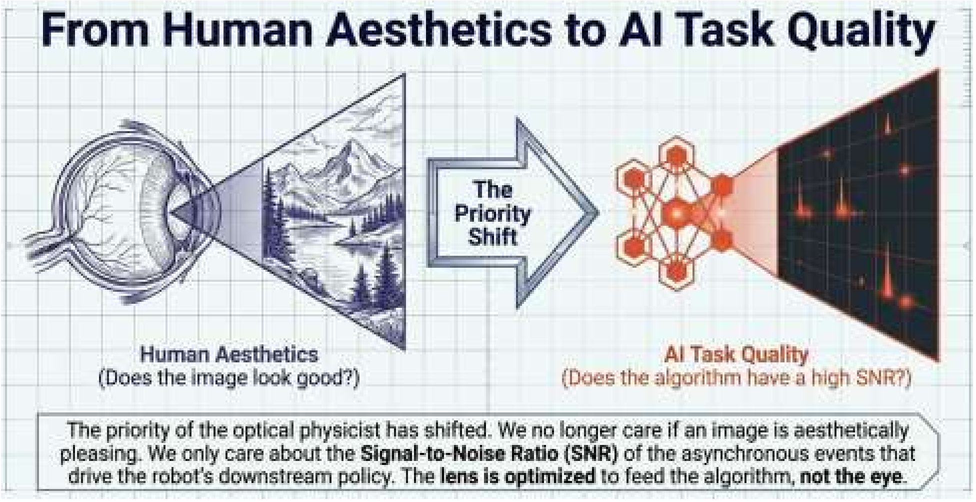


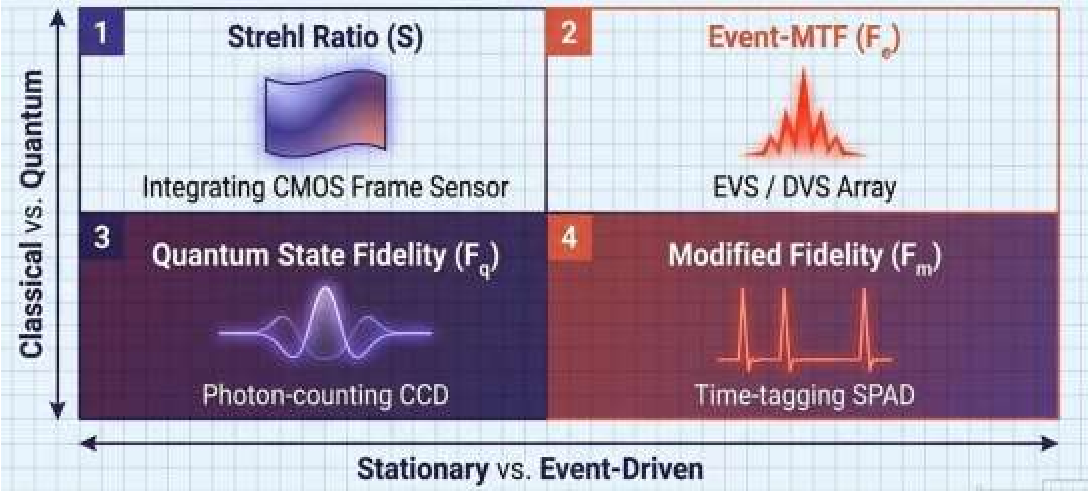
Core theme

When the detector changes, the lens merit function must change with it.

Background – Why we consider these







Background - Why Strehl/ Kullback-Leibler

Strehl:

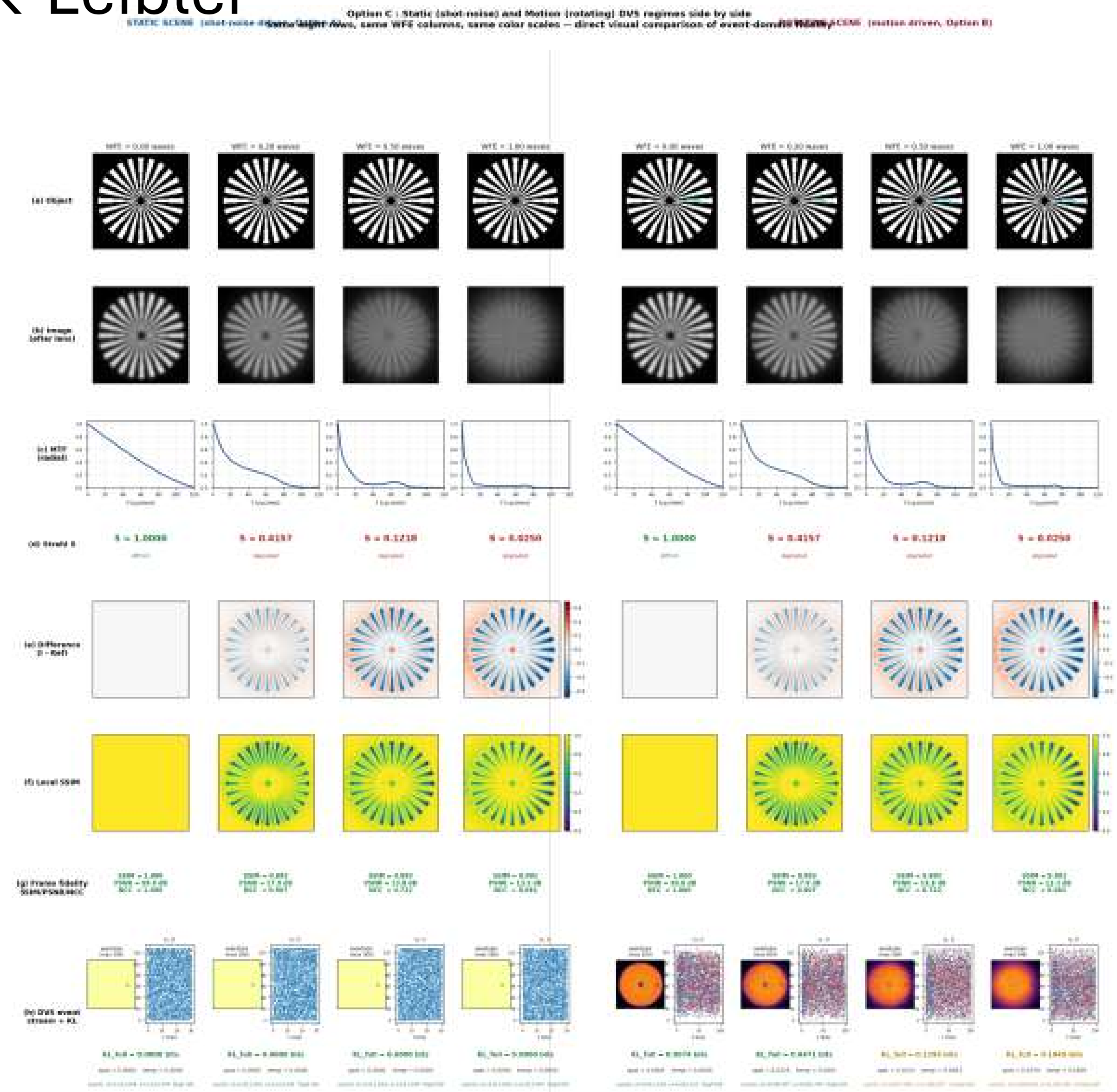
The Strehl ratio $S = \max |h(\xi, \eta)|^2 / \max |h_{diff}(\xi, \eta)|^2$ is the peak-PSF figure of merit for the ***stationary, photon-rich, integrating-detector*** regime and is governed in the small-error limit by Maréchal's expansion $S \approx \exp\left[-(2\pi\sigma_w)^2\right]$.

Strehl ratio: widely used in optics

Kullback–Leibler:

The Kullback–Leibler divergence $D_{\mathrm{KL}}(P_{ideal} \parallel P_{actual})$, evaluated on the photon-arrival / event-time distribution $P(t)$ is the natural fidelity for event-driven and single-photon regimes. Wrapped as $F_m = \exp(-D_{\mathrm{KL}})$ it lives in $[0, 1]$ alongside S, but unlike S, it is sensitive to the ***full shape*** of the arrival distribution, including the tails that amplitude-overlap measures discard.

1951 paper
≈75 years of history, with deeper roots



object

image

Characterization from classical to quantum

Strehl's history is intra-optical*, growing from the diffraction theory of telescopes;* ***KL's history is extra-optical****, arriving in optics only after detours through Bell Labs, the NSA, statistical mechanics, von Neumann algebras, and finally machine learning.*

The design target moved: from image quality to event quality

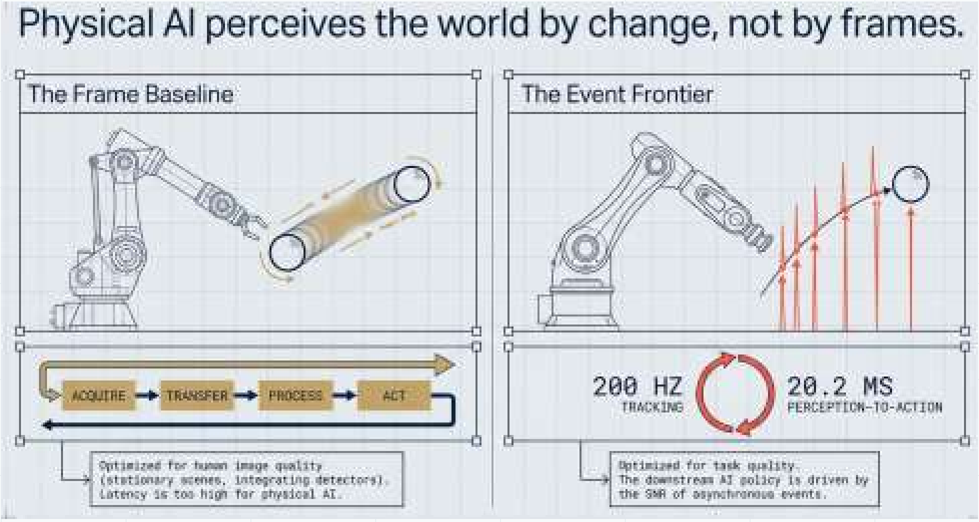
Physical-AI systems perceive change; the optic must preserve the right signal.

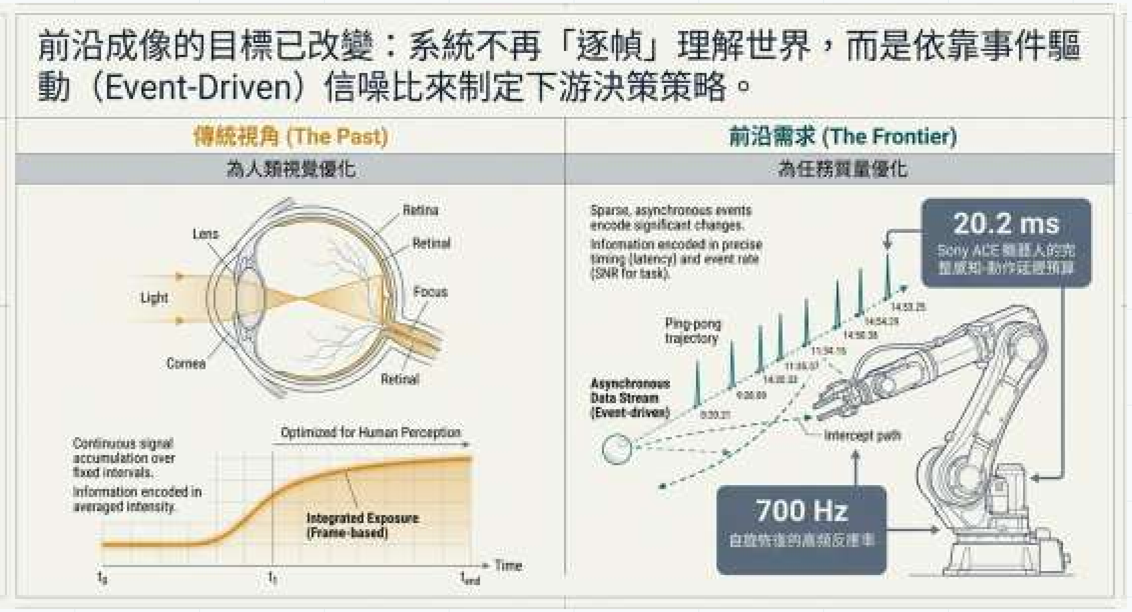
Old lens-design default

- Optimize for stationary image fidelity
- Use Strehl / MTF as scalar quality metrics
- Treat latency and policy performance as downstream issues

→ New frontier

- Optimize for the SNR of useful events
- Keep moving-edge contrast, timing, and tails intact
- Make stray light a first-order event-noise source





Same scene, two measurement protocols

A moving edge is a simple but decisive test.

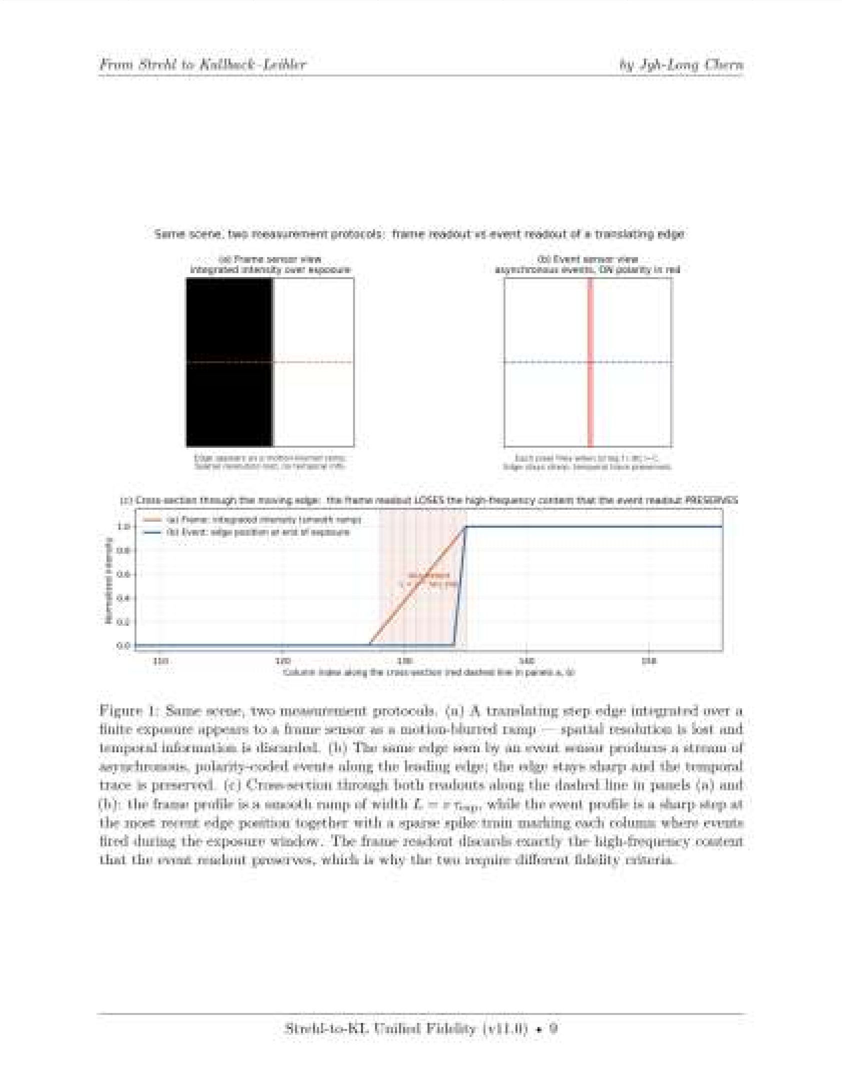

From Strehl to Kullback-Leibler — *by Jyh-Long Chern*

Figure 1: Same scene, two measurement protocols. (a) A translating step edge integrated over a finite exposure appears to a frame sensor as a motion-blurred ramp — spatial resolution is lost and temporal information is discarded. (b) The same edge seen by an event sensor produces a stream of asynchronous, polarity-coded events along the leading edge; the edge stays sharp and the temporal trace is preserved. (c) Cross-section through both readouts along the dashed line in panels (a) and (b): the frame profile is a smooth ramp of width $L = v\,\tau_{exp}$, while the event profile is a sharp step at the most recent edge position together with a sparse spike train marking each column where events fired during the exposure window. The frame readout discards exactly the high-frequency content that the event readout preserves, which is why the two require different fidelity criteria.

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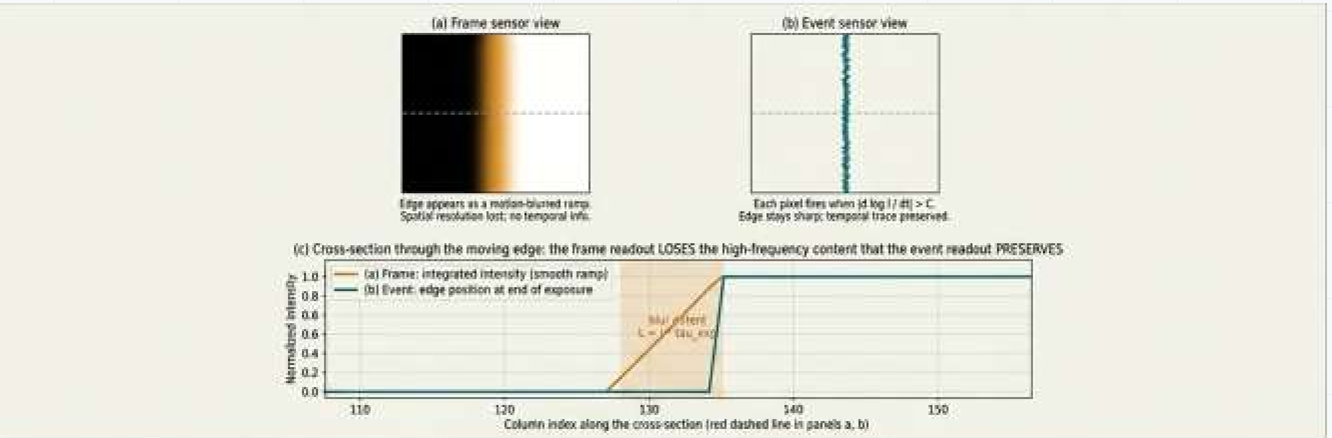


傳統指標的盲區	真實的物理測量	結論
斯特列爾比 (Strehl ratio) 評估的是累積強度的空間峰值 (I)。	事件傳感器測量的是對數強度的時間導數 $(\partial \log I / \partial t)$。	Strehl 評估了一個事件傳感器根本不測量的物理量。

Frame sensor

- Integrates intensity over exposure.
- A moving edge becomes a ramp; high spatial frequency is averaged away.

Event sensor

- Fires when local log-intensity changes.
- The same edge becomes asynchronous spikes with timing information.

Design implication

A PSF-peak metric and a gradient/timing metric are not measuring the same physical object.

Why Strehl fails for event optics

Strehl measures peak PSF intensity; event cameras measure ∂logI/∂t.

Strehl ratio S

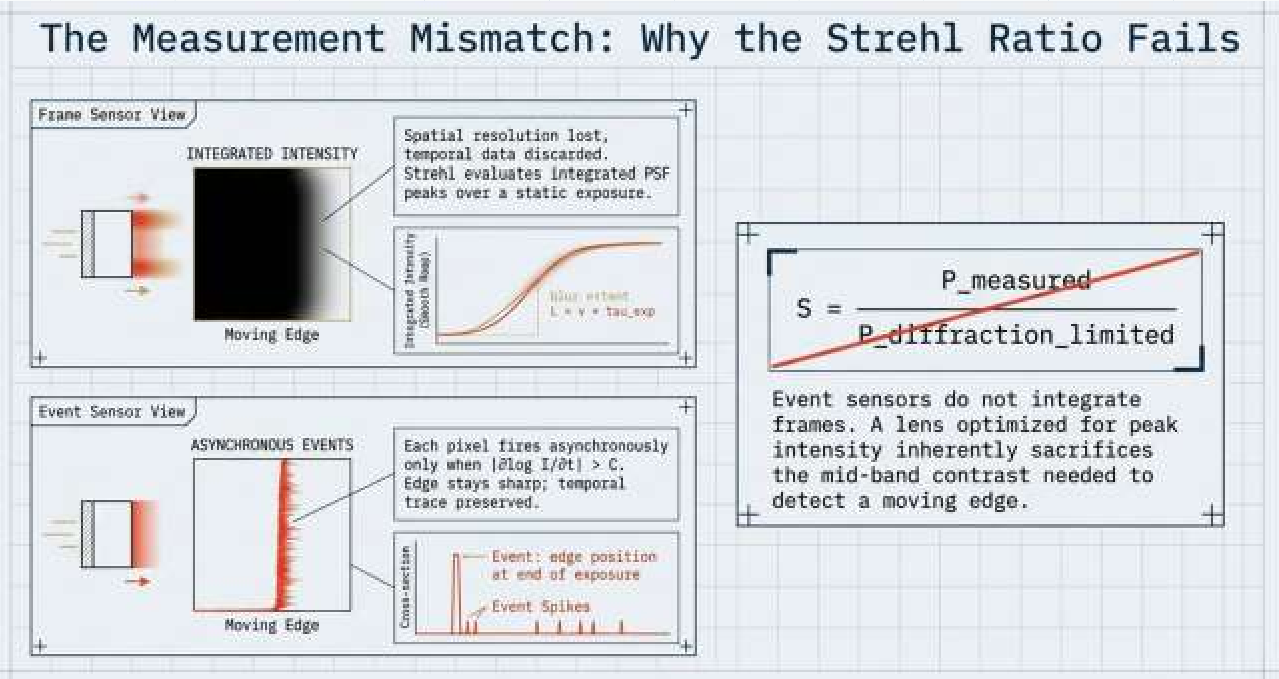
S = peak PSF / diffraction peak

Best scalar when scene is stationary and detector integrates frames.

Event response

trigger when |∂logI/∂t| > C/Δt

Best scalar must follow edge velocity, gradient, contrast threshold, and event noise.

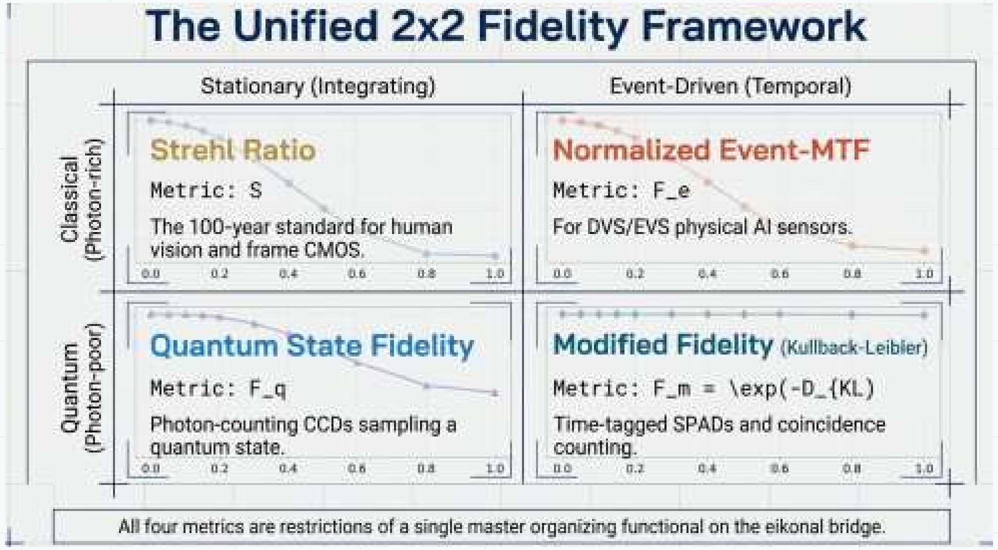


Therefore:

A Strehl-only merit can reject lenses that are strong event-optics candidates, and accept lenses that create false events.

The 2×2 fidelity map: detector protocol × photon statistics

Four useful scalar metrics, not one universal scalar.



C–S
Strehl S
Classical / stationary / integrating CMOS.

C–E
Event-MTF Fe
Classical / change-driven / DVS-EVS.

Q–S
Fidelity Fq
Photon-poor state or coherent mode overlap.

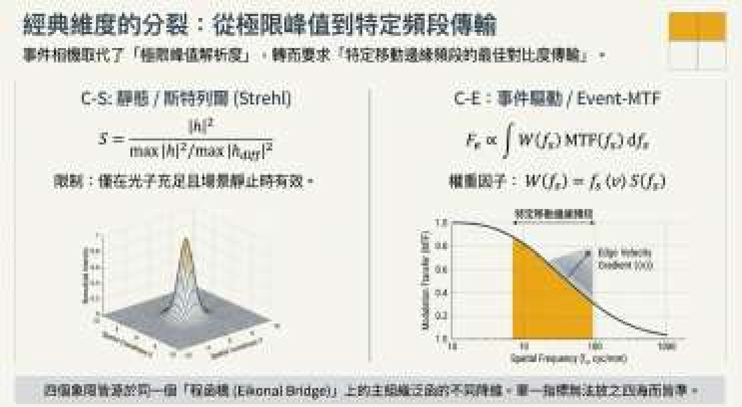
Q–E
exp(−DKL)
Time-tagged photon streams and coincidences.

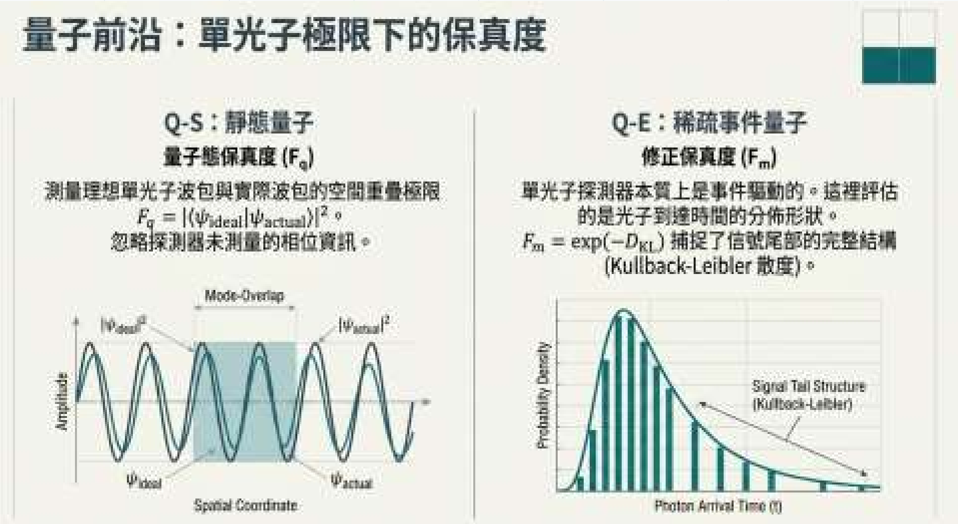
Useful reading rule

- Rows tell photon statistics.
- Columns tell detector protocol.
- The correct metric lives in the cell of your hardware.

Four metrics, one organizing idea

Each metric is a restriction of the same measurement-aware fidelity functional.





C–S: Strehl

$S = \max|h|^2 / \max|hDL|^2$

Use when peak PSF intensity predicts system value.

Q–S: Quantum state fidelity

$Fq = |\langle \psi ideal \mid \psi actual \rangle|^2$

Use for coherent projection / mode-overlap receivers.

C–E: Normalized Event-MTF

$Fe = \int W(fs)MTF(fs)dfs / \int W(fs)MTFDL(fs)dfs$

Use when moving-edge contrast and timing drive the policy.

Q–E: KL modified fidelity

$Fm = \exp[-DKL(Pideal \,||\, Pactual)]$

Use when the output is an arrival-time or coincidence distribution.

Master design habit

Define M as the detector measurement statistic first.
Only then choose the fidelity functional F(Mideal, Mactual).

Numerical evidence: different metrics decay differently

The disagreement away from the diffraction limit is the practical signal.

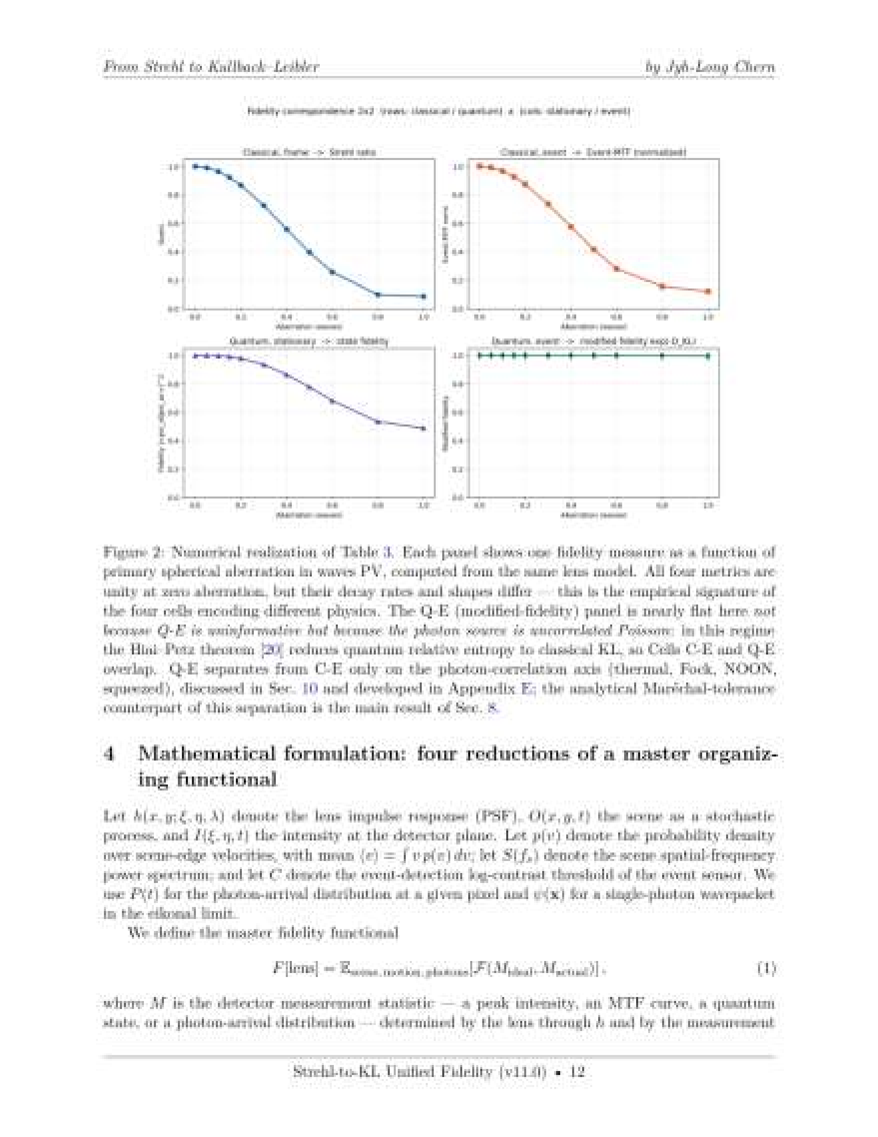


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Figure 2: Numerical realization of Table 3. Each panel shows one fidelity measure as a function of primary spherical aberration in waves PV, computed from the same lens model. All four metrics are unity at zero aberration, but their decay rates and shapes differ — this is the empirical signature of the four cells encoding different physics. The Q-E (modified-fidelity) panel is nearly flat here *not because Q-E is uninformative but because the photon source is uncorrelated Poisson*: in this regime the Hiai–Petz theorem [20] reduces quantum relative entropy to classical KL, so Cells C-E and Q-E overlap. Q-E separates from C-E only on the photon-correlation axis (thermal, Fock, NOON, squeezed), discussed in Sec. 10 and developed in Appendix E; the analytical Maréchal-tolerance counterpart of this separation is the main result of Sec. 8.

4 Mathematical formulation: four reductions of a master organizing functional

Let $h(x, y; \xi, \eta, \lambda)$ denote the lens impulse response (PSF), $O(x, y, t)$ the scene as a stochastic process, and $I(\xi, \eta, t)$ the intensity at the detector plane. Let $p(v)$ denote the probability density over scene-edge velocities, with mean $\langle v \rangle = \int v\, p(v)\, dv$; let $S(f_s)$ denote the scene spatial-frequency power spectrum; and let C denote the event-detection log-contrast threshold of the event sensor. We use $P(t)$ for the photon-arrival distribution at a given pixel and $\psi(\mathbf{x})$ for a single-photon wavepacket in the eikonal limit.

We define the master fidelity functional

$$F[\text{lens}] = \mathbb{E}_{\text{scene, motion, photons}}[\mathcal{F}(M_{\text{ideal}}, M_{\text{actual}})], \tag{1}$$

where M is the detector measurement statistic — a peak intensity, an MTF curve, a quantum state, or a photon-arrival distribution — determined by the lens through h and by the measurement

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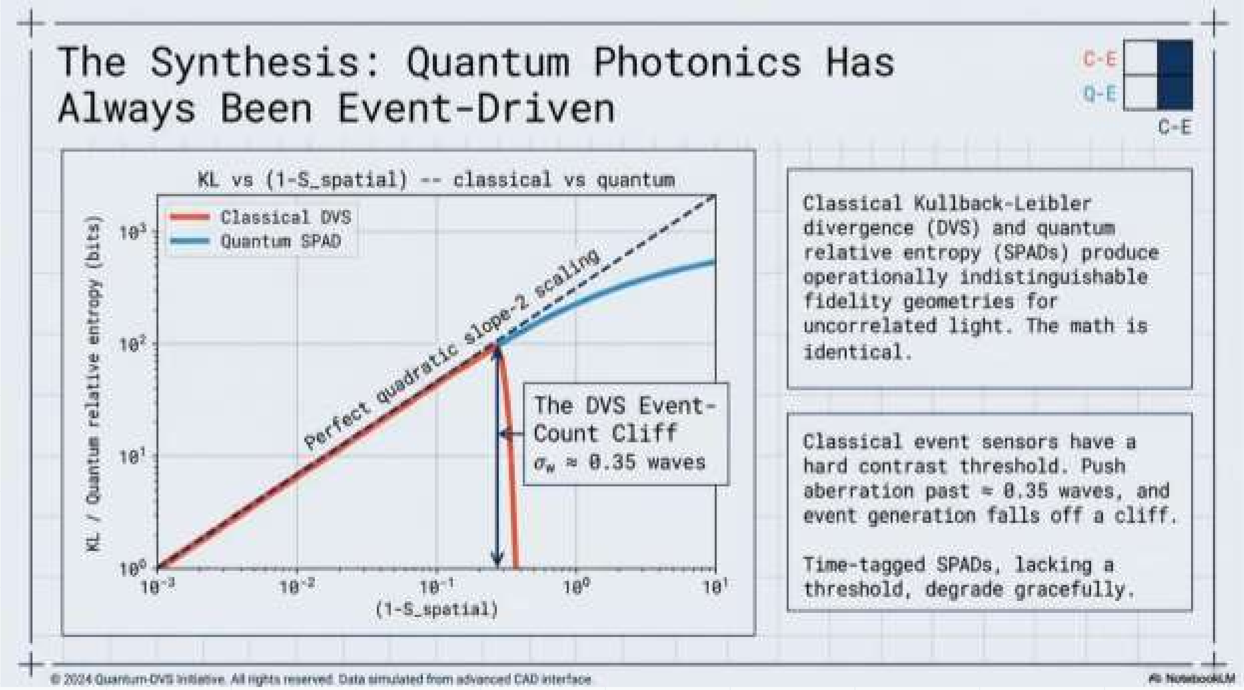
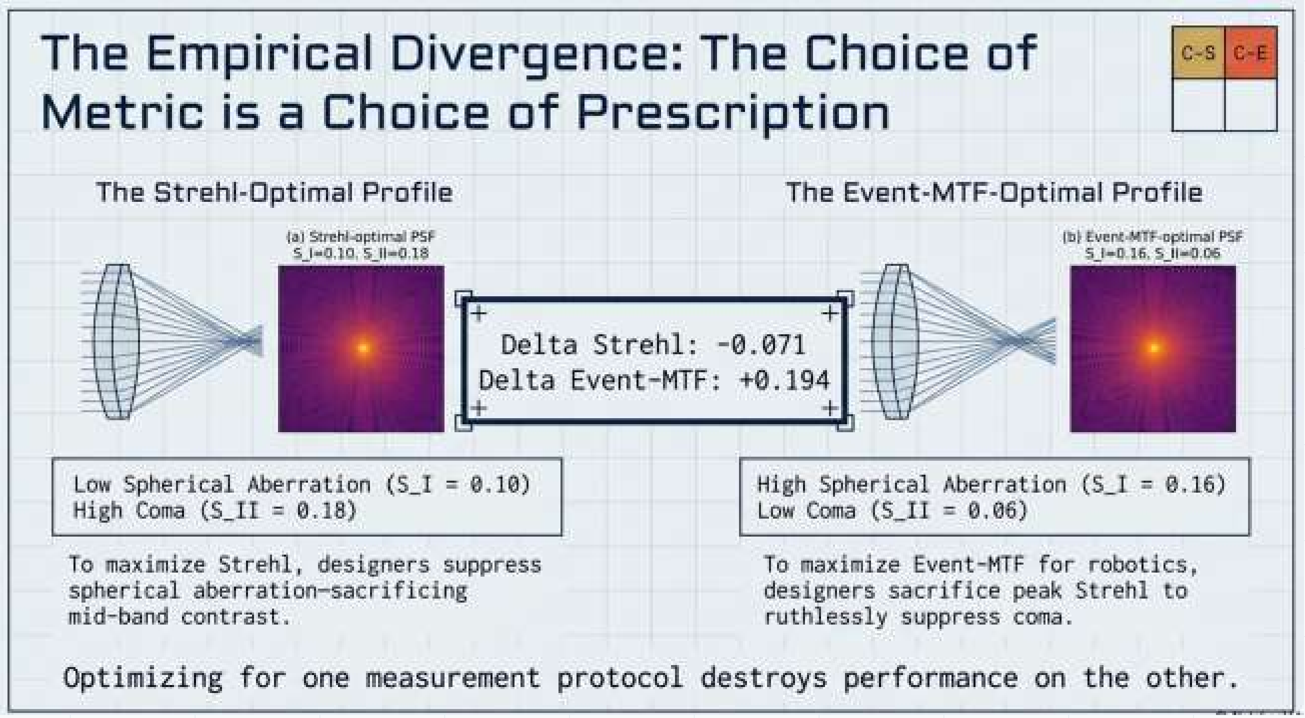


Engineering reading

At zero aberration all metrics agree.
Away from zero, they rank the lens differently.
That means they drive different optimization directions.

Metric choice becomes prescription choice

Optimizing for Strehl and optimizing for event-MTF can produce different lenses.



Strehl-optimal profile

Suppresses peak-blurring aberrations.
May sacrifice mid-band contrast for moving edges.

Event-MTF-optimal profile

Protects edge-gradient transfer.
May accept lower peak Strehl if task SNR improves.

Practical implication

Do not start event-optics design by optimizing Strehl and “checking” event performance later.

Practical reset for working lens designers

The new priority stack is not cosmetic; it changes tolerance and cost.

FOM	Frame baseline	Event starting target
PST	10^{-3}–10^{-4}	$\leq 10^{-5}$
Per-surface R	~0.5% AR	≤0.1%
CRA	derived	primary variable
Assembly QC	visual / CMOS	photon-counting

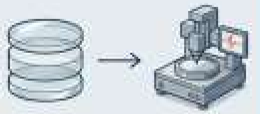
Cost inversion

- Visible frame optics: glass count often dominates.
- Event/NIR optics: coating, surface finish, baffling, and QC can dominate.

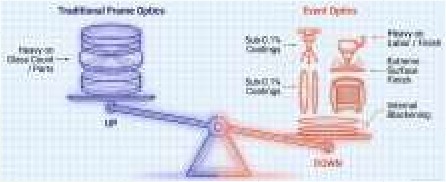
The Practical Reset: A New Playbook for Engineers

Parameter	Frame-Optic Baseline (Old Way)	Event-Optic Target (New Way)
Stray Light (PST)	10^{-3} to 10^{-4}	$\leq 10^{-5}$
Per-Surface Reflectance	~ 0.5% AR	< 0.1%
Chief-Ray Angle (CRA)	Derived constraint	Primary design variable
Quality Control	Visual / CMOS Bench	Photon-Counting Assembly QC

BOM Economics Invert



These new fidelity metrics demand a structural inversion of manufacturing. In the visible and NIR space, glass count no longer dominates cost. Because of how event sensors respond to stray light, cost is now dominated by surface finish, deep-well CRA control, and 0.1% coating tolerances.



工作設計師的實用重置 (The Practical Reset)

成本邏輯發生倒轉。BOM 成本從「玻璃鏡片數量」轉移到「表面光潔度與鍍膜勞動力」。

規格參數	傳統幀相機基準 (Frame Baseline)	事件相機新目標 (Event Target)
PST (點光源透過率)	10^-3 ~ 10^-4	≲ 10^-5 (避免鬼影引發假事件)
AR 鍍膜反射率	~ 0.5%	≲ 0.1% 每面 (動態鬼影控制)
主光線角 (CRA)	衍生被動約束	首要設計變量 (深阱像素架構需求)
組裝 QC (質量控制)	視覺 / 常規檢查	光子計數級 QC (次閾值缺陷足以使傳感器飽和)

BOM Economics

成本邏輯發生倒轉。BOM 成本從「玻璃鏡片數量」轉移到「表面光潔度與鍍膜勞動力」。

The ghost-event rate budget: why stray light becomes first-order

In event optics, a ghost is not just glare — it can be a deterministic false feature.

PST
out-of-field source leaks in

R
coating bounce occurs

ω, CRA
ghost moves across field

C, τref
pixels fire at threshold

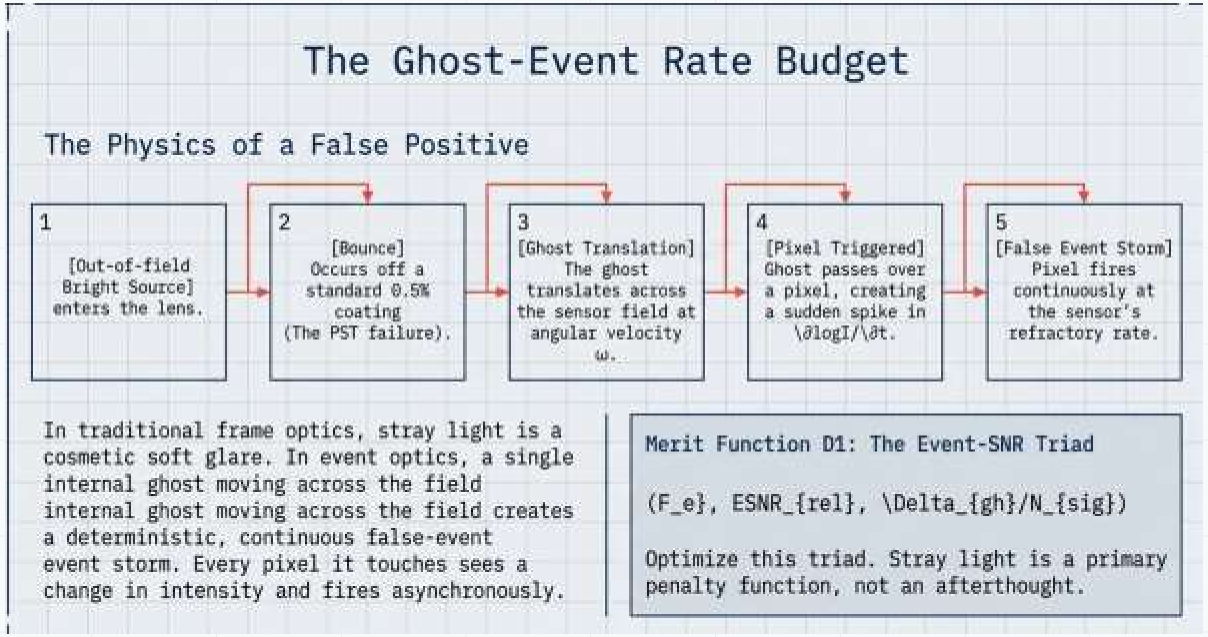
Ghost count enters Event-SNR

ESNR = Nsig / √(Nsig + Ndark + Ngh)

Structured ghost terms behave like bias, not just variance.

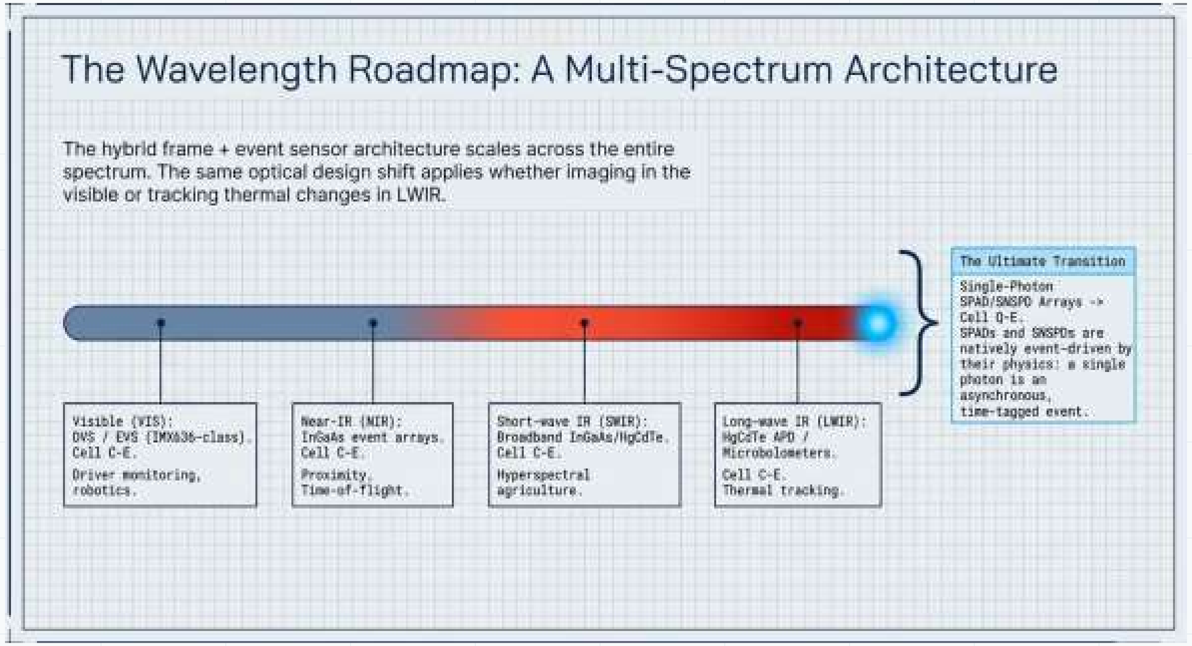
Design rule

Budget PST, coating reflectance, internal baffling, CRA, and contrast threshold together — not as separate line items.



Roadmap: one architecture across spectrum

The same detector-protocol logic scales from visible event cameras to single photons.



Application domain 1: Visible
DVS / EVS
robotics, monitoring

Application domain 2: NIR
InGaAs / ToF
proximity, ranging

Application domain 3: SWIR
InGaAs
hyperspectral, aperture

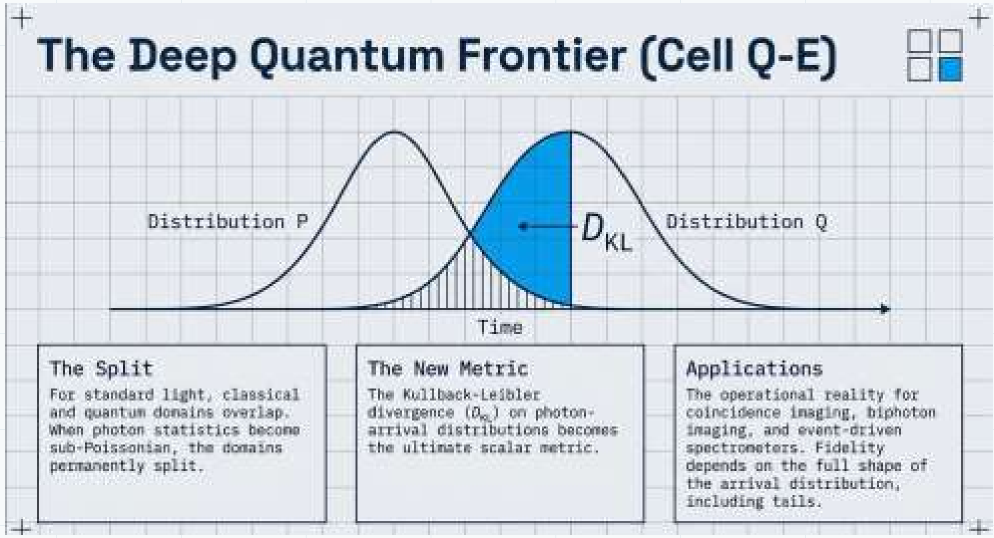
Application domain 4: LWIR
HgCdTe
thermal tracking

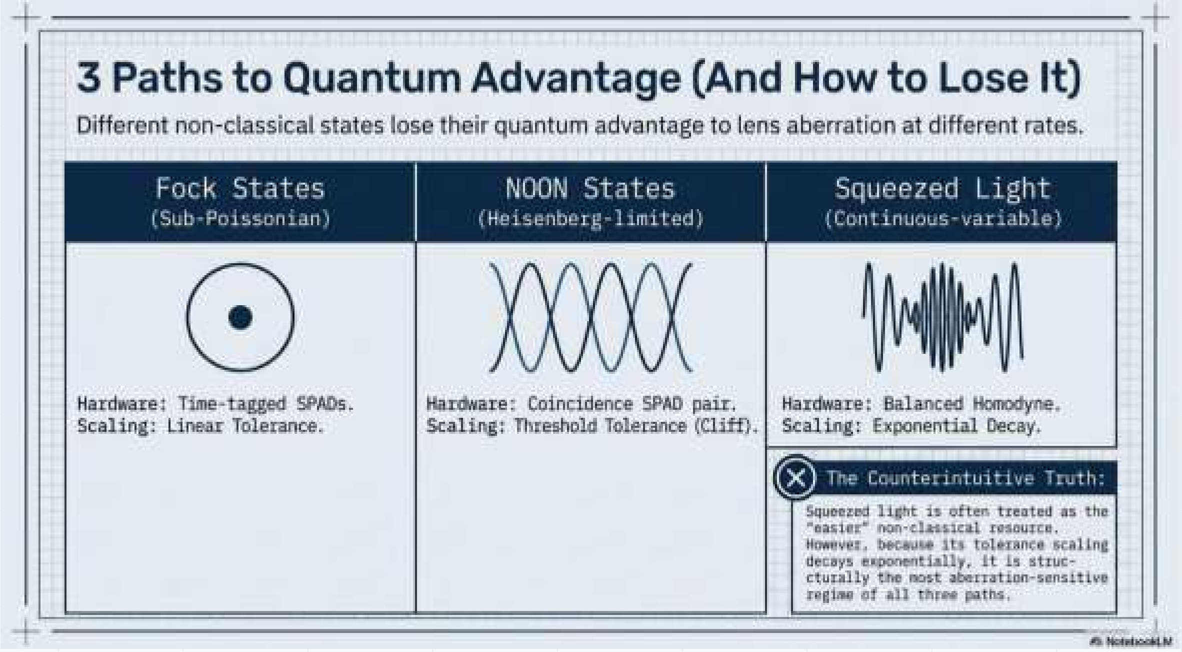
Application domain 5: Single photon
SPAD / SNSPD
arrival-time optics

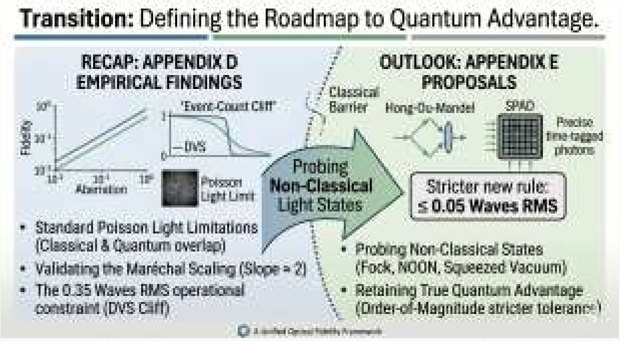
Hybrid frame + event readout becomes an optical architecture, not only a sensor choice.

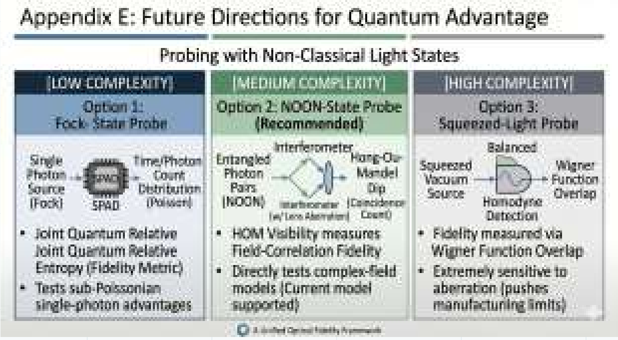
Quantum photonics was event-driven from the start

A photon arrival is already an asynchronous time-tagged event.









Design transition

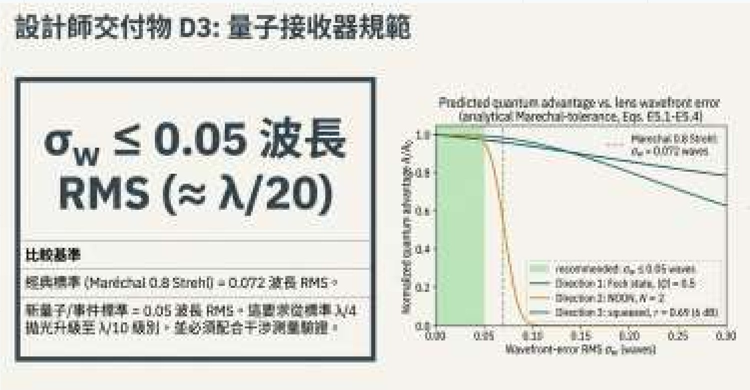
- For photon-rich frames: use amplitude/PSF scalars.
- For time-tagged photons: use distributional fidelity on P(t).

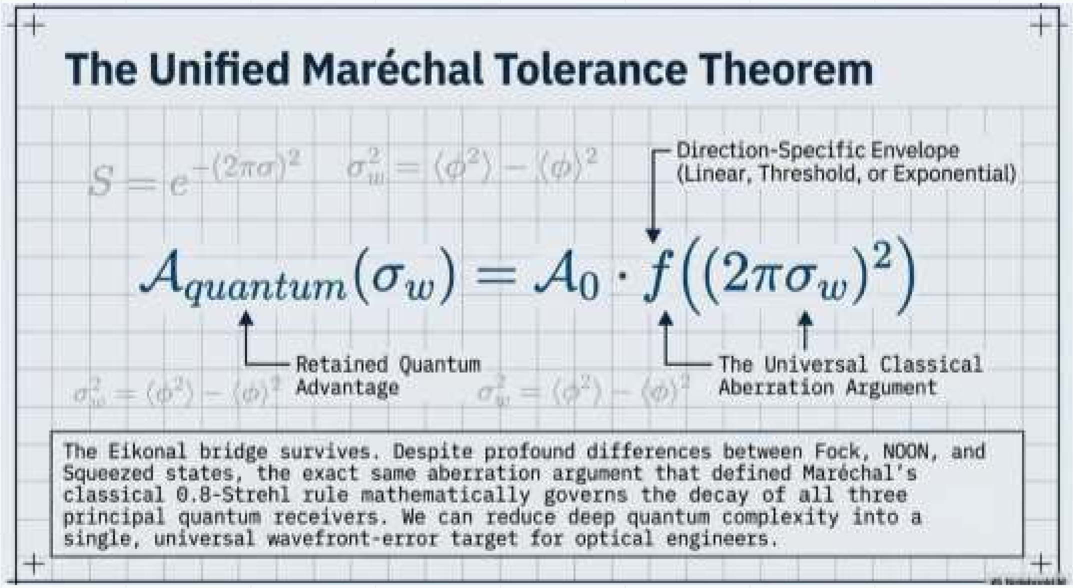
Warning

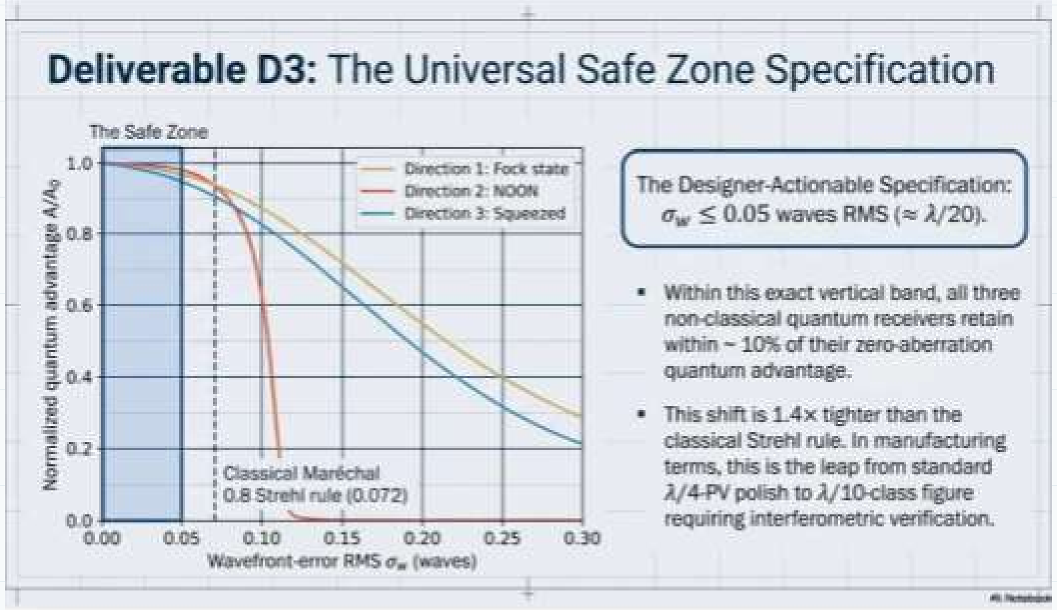
Fock, NOON, and squeezed-light receivers lose advantage differently — the lens tolerance must be state-aware.

Unified Maréchal tolerance: from classic 0.8 Strehl to quantum safe zone

A practical specification emerges: σw ≤ 0.05 waves RMS.







Classical Maréchal

S ≈ exp[−(2πσw)^2]

Common diffraction-limited threshold: Strehl ≈ 0.8 at σw ≈ 0.072 waves.

Quantum/event safe target

σw ≤ 0.05 waves RMS

A tighter working target for retained quantum advantage across receiver families.

Implementation: new operands, not new physics

CODE V, Zemax, LightTools, and RSoft can host the framework today.

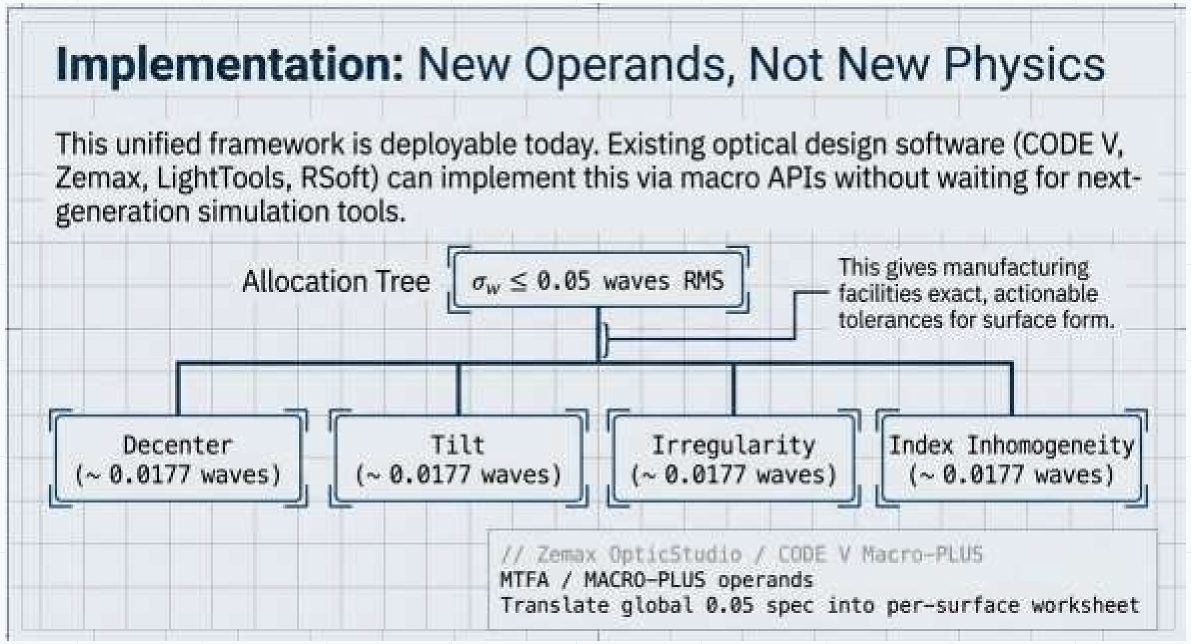
1. Choose cell C–S / C–E / Q–S / Q–E **(see p.5)** → 2. Build merit S, Fe, Fq, exp(−DKL) + ESNR → 3. Allocate tolerances WFE → decenter / tilt / irregularity / Δn

CODE V / Zemax
Add user operands or macros for Event-MTF, ESNR, and KL-related post-processing.

LightTools
Use stray-light paths and PST to feed the ghost-event budget.

RSoft / photonics
Map field propagation into arrival distributions, modes, and coupling metrics.

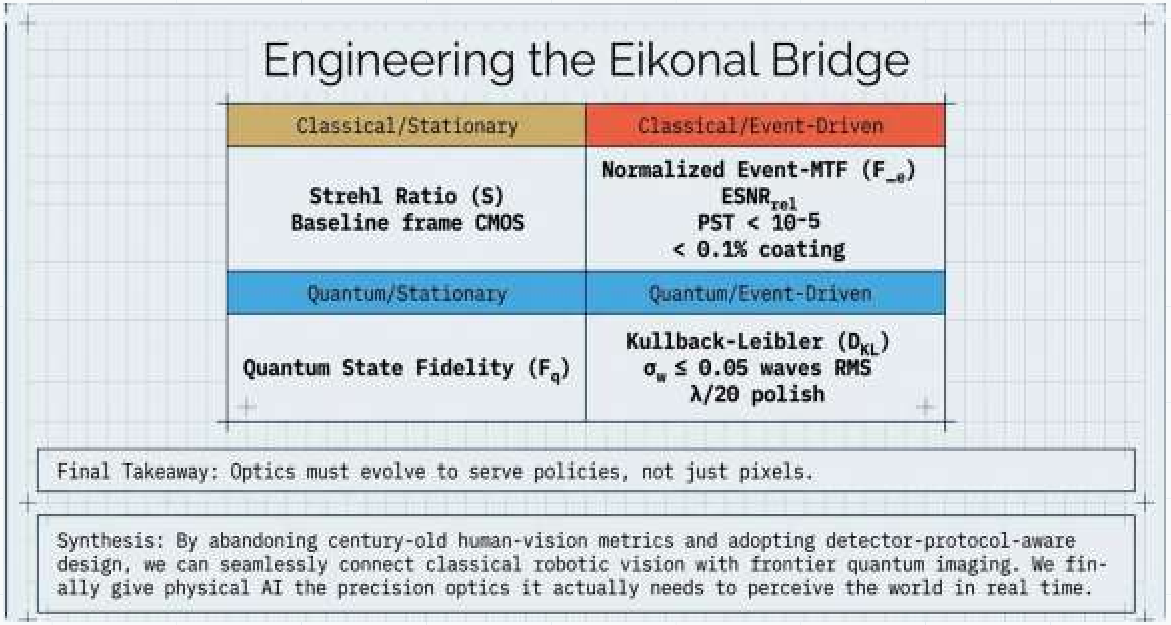
QC loop
Photon-counting or event-sensor bench validates false-event rate, not just image quality.



Closing playbook for optical engineers

Make the detector protocol explicit before optimization begins.

Question	Engineering answer
What does the detector measure?	peak intensity, MTF band, mode overlap, or event distribution
What is the task signal?	human image quality, moving-edge SNR, photon-arrival fidelity, or coincidence shape
What is the false-positive source?	blur, ghost event, dark burst, timing tail, or state decoherence
What must be toleranced?	WFE plus PST, R, CRA, baffling, and assembly QC



Final takeaway

Optics must evolve to serve policies, not just pixels.

Open questions

From Strehl to Kullback–Leibler *by Jyh-Long Chern*

9 Open questions

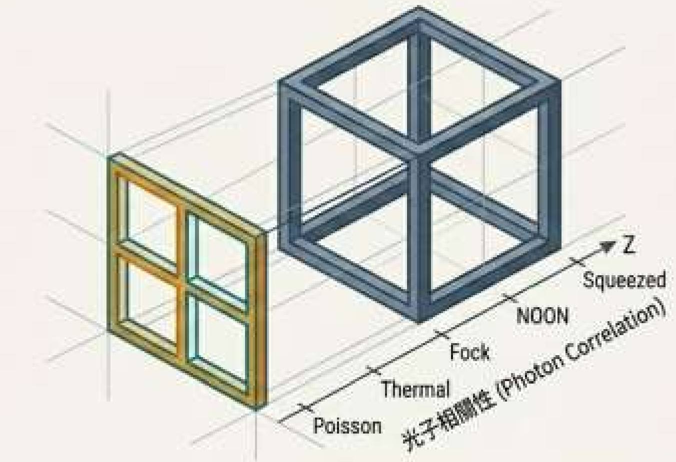
Five questions remain for future work. Table 7 summarizes them; each is briefly motivated below.

Table 7: Summary of open questions, with the reason each is genuinely open today and a likely approach for resolution.

#	Question	Why open	Likely approach
1	Closed form for F_e vs S as a function of aberration type	Empirical $\beta \in [0.5, 0.7]$ depends on Zernike content; no analytical reduction exists	Zernike-decomposed analytical MTF integral with mid-band weighting
2	Polarization-resolved extension of the 2×2	Polarization adds two channels per measurement; current framework is scalar-only	Lift to a Stokes-vector fidelity; preserve cell structure with vector-valued M
3	Tolerance behavior of F_e vs S	Standard tolerancing tables are derived for Strehl; behavior under Event-MTF is empirically untested [19]; Appendix F gives a partial answer for the quantum-receiver column	Monte Carlo over decenter / tilt / surface-form distributions with the merit set to $(F_e, \mathrm{ESNR}_{\mathrm{ed}})$
4	Information-theoretic unification of Eq. (1)	Four cells are conceptually unified but not algebraically; mutual information is a candidate	Express $\mathcal{F}$ as Bayesian regret over decoders; recover four cells as limits
5	Photon-correlation axis ($2 \times 2 \times 2$ promotion)	Photon correlation is a third axis not captured by current framework; Hiai–Petz collapse only holds for diagonal photon-number distributions	Add a coherence axis; recover Hong–Ou–Mandel fidelity [11] in one octant and $g^{(2)}$ in another. Sec. 8 provides the analytical scaffolding; Appendix E the experimental directions

未來視野：從 2x2 到 2x2x2 的維度躍遷

光學設計已正式進入探測器協議感知（Detector-Protocol-Aware）的全新維度。



物理 AI 不僅是機器人技術的故事；它是跨越所有波長、深入量子領域的光學路線圖。當我們超越無相關性的泊松光（Poisson light）時，經典與量子事件保真度的差異將完全顯現。

Q5. Photon-correlation axis. Photon correlations introduce a third axis (uncorrelated Poisson, thermal, Fock, NOON, squeezed). The 2x2 likely promotes to 2x2x2, with Hong–Ou–Mandel-class fidelity [11] in the coherent-quantum-event octant and Glauber g(2) in the thermal-quantum-event octant. Section 8 provides the analytical scaffolding for the three principal non-classical directions and gives the unified Maréchal-tolerance theorem [Eq. (15)]; Appendix E develops the corresponding experimental directions